

Sample End Semester Question Paper - 2025

Subject: Theory of Computation | Code: CS303 | Semester: 3

Branch: CSE | Regulation: R20 (sample) | Time: 3 Hours | Max Marks:

This is a SAMPLE question paper generated for Backlog Buddy (demo/testing data).
It is not an official university paper and is provided to demonstrate the platform.

PART A - Short questions (10 x 1 = 10 Marks)

1. What is CNF?
2. State the pumping lemma for regular languages in words.
3. What is a parse tree?
4. Define a terminal symbol.
5. What is a decider?
6. Define a finite automaton.
7. State the difference between a DFA and an NFA.
8. What is a regular expression?
9. What is a grammar?
10. Name the two types of derivation in a CFG.

PART B - Long questions (8 x 10 = 80 Marks)

1. Explain context-free grammars: leftmost and rightmost derivations, parse trees.
2. Convert a given CFG into Chomsky normal form.
3. Apply the pumping lemma to show that a given language is not regular.
4. Explain the Chomsky hierarchy of languages and grammars.
5. Construct a DFA/NFA for a given language with a state diagram.
6. Convert an NFA into a DFA with a complete example.
7. Explain regular languages and regular expressions. Prove closure properties.
8. Minimise a given DFA using the partition method.